

The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

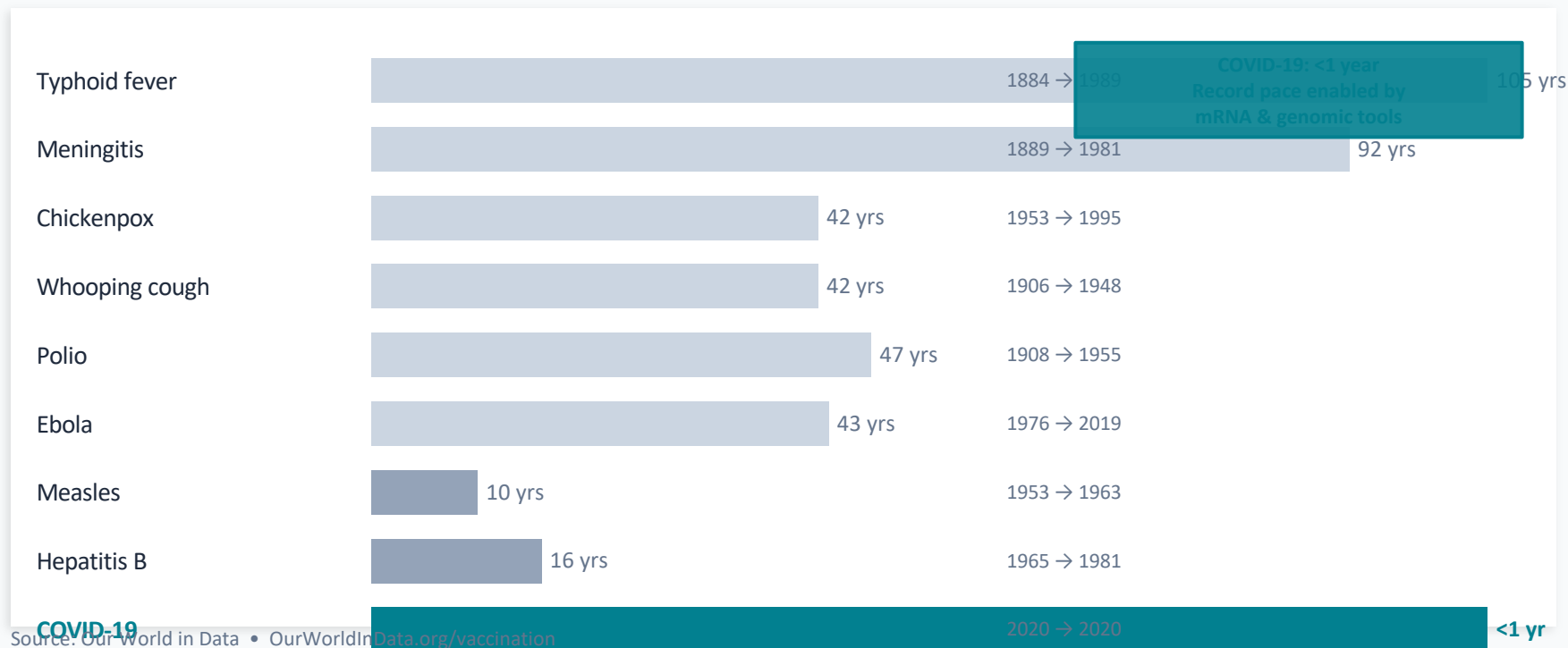
A data-driven overview of vaccine impact across two centuries

1796 – 2020

From 100 Years to Under 1 Year

Vaccine Development Is Accelerating

Years from pathogen identification to vaccine licensure



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

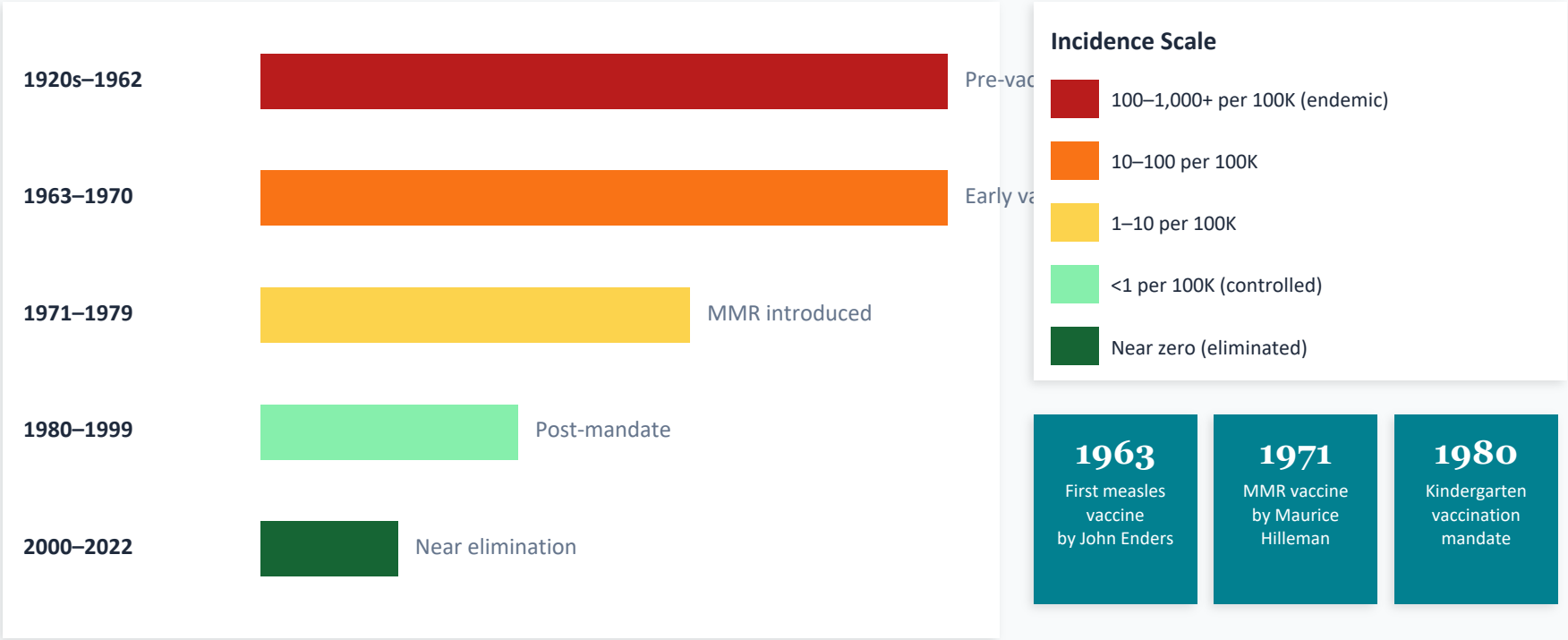
Pre-vaccine vs. post-vaccine annual case and death rates per million population, US

Disease	Pre-Vaccine Cases /M	Post-Vaccine Cases /M	Case Reduction	Pre-Vaccine Deaths /M	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hep. B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

= 100% elimination of cases and deaths

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles incidence per 100,000 population, 1929–2022



Before 1963: nearly every state reported 100–1,000+ cases per 100,000. After 1980 mandate: cases near zero across all states, with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination

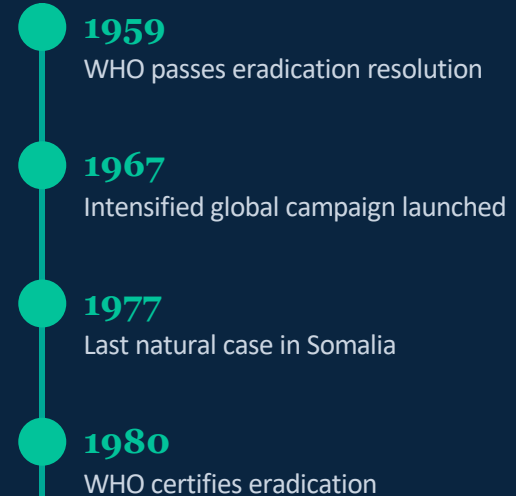
Why Smallpox Could Be Eradicated

- 1 The virus infected only humans — no animal reservoir
- 2 Symptoms were visible and distinguishable
- 3 The vaccine was cheap, stable, and highly effective
- 4 Infected individuals could be traced and isolated

Global Eradication Campaign

WHO passed a resolution in 1959. An intensified campaign launched in 1967. The last natural case occurred in Somalia in 1977. Certification of eradication came in December 1980.

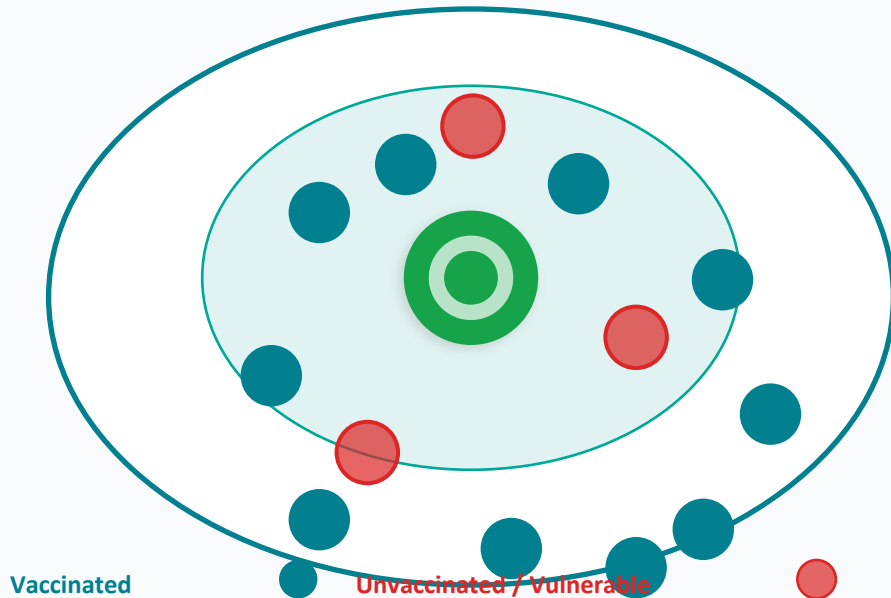
Eradication Timeline

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- 1959**
WHO passes eradication resolution
 - 1967**
Intensified global campaign launched
 - 1977**
Last natural case in Somalia
 - 1980**
WHO certifies eradication

The only human disease ever fully eradicated

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Population-level vaccination shields those who cannot be vaccinated



Who Depends on Herd Immunity?

Newborns

Too young for certain vaccines

Immunocompromised

Cannot mount adequate immune response

Chemotherapy patients

Immune systems weakened by cancer treatment

Allergic individuals

Cannot receive certain vaccine components

When vaccination coverage is high enough, pathogens cannot spread effectively — protecting entire communities, including those who cannot be vaccinated.

Gaps Persist: Diseases Where Vaccine Impact Is Still Incomplete

Pneumococcal disease, hepatitis A and B still show significant remaining burden in the US

Pneumococcal Disease

40.5%

case reduction achieved

59.5% gap remaining

Death reduction: 31.3%

Barriers to full impact:

- Serotype coverage limitations
- Access in underserved communities
- Cost and supply chain barriers

Hepatitis A

89%

case reduction achieved

11% gap remaining

Death reduction: 88.7%

Barriers to full impact:

- Outbreak response gaps
- Homelessness and transient populations
- Variable state vaccination mandates

Acute Hepatitis B

83.9%

case reduction achieved

16.1% gap remaining

Death reduction: 83.6%

Barriers to full impact:

- Testing and diagnosis delays
- Stigma limiting uptake
- Birth dose coverage gaps

Globally, not everyone has access to vaccines. Supply shortages and declining trust put children and communities at risk.

Key Takeaways

Vaccines have eliminated deadly diseases — but the job isn't done

01 100% Elimination Achieved

Diphtheria, smallpox, and both forms of polio have been eliminated in the US through vaccination — zero cases, zero deaths.

02 Development Timelines Collapsed

From 105 years (typhoid) to under 1 year (COVID-19) — mRNA technology and genomic tools have revolutionized vaccine development speed.

03 Herd Immunity Protects Communities

When enough people are vaccinated, pathogens cannot spread effectively — shielding newborns, immunocompromised patients, and chemotherapy patients.

04 Gaps Persist for Some Diseases

Pneumococcal disease (40.5% reduction), hepatitis A (89%), and hepatitis B (83.9%) show incomplete control — access, coverage, and trust challenges remain.

Call to Action: Improve global access, strengthen coverage, and build trust in vaccination to protect every community.